LABORATORY PROTOCOL · TSM2370 MATERIALTEKNIKK

Lab 1: Axial Tensile Testing and Strain Measurement Accuracy

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Experiment date	*[DD/MM/YYYY]*
Lab partner(s)	*[Lab partner names]*

1. Objective

Determine how an unknown aluminium alloy behaves under a controlled, axial pulling load until fracture. The experiment also demonstrates why an extensometer is needed, instead of the machine's crosshead travel, to calculate strain accurately.

2. Theoretical background

Engineering stress (σ) is the applied axial force (F) divided by the original cross-sectional area (A_0). Engineering strain (ε) is the change in length (ΔL) divided by the original gauge length (L_0). The load cell of the universal testing machine (UTM) measures force accurately, but strain calculated from crosshead travel includes errors from the grips and the machine frame. An extensometer measures the elongation directly over the gauge length. The material goes from linear elastic deformation into plastic deformation, with strain hardening, necking and finally fracture.

2.1 Relevant formulas

Engineering stress: $\sigma = F / A_0$

Engineering strain: $\varepsilon = \Delta L / L_0$

Young's modulus: $E = \sigma / \varepsilon$ *(linear elastic region)*

3. Materials and equipment

Device / material	Amount / specification	Manufacturer	Notes
Universal testing machine	10 kN (max. 5 kN grips)	Lloyd (LR 10K)	Axial pulling
Camera extensometer	1 unit	Basler / Logitech	Optical strain measurement
Aluminium sheet dog-bones	1 mm thick, 12.5 mm wide	ISO standard	Unknown alloy test coupons
Digital calipers	1 unit	–	For L_0, width and thickness

4. Experimental procedure

4.1 Experimental setup

The Lloyd LR 10K is fitted with tensile grips. The camera extensometer is mounted facing the specimen and tracks optical strain markers on the gauge section. The aluminium dog-bone is centred in the grips.

4.2 Procedure

1. Take safety precautions and make sure the UTM safety shields are in place.
2. Use digital calipers to measure the initial gauge length (L_0) and the width and thickness of the stamped aluminium part.
3. Enter the measured L_0 in the extensometer software and clamp the specimen securely in the UTM grips.
4. Start the tensile pull at the specified crosshead speed.
5. Run the test until the specimen fractures completely.
6. Export the raw force and elongation data from the LR 10K software.
7. Clean the workstation and dispose of the fractured aluminium pieces safely.

5. Results and measurements

Parameter / measurement	Value	Unit
Yield strength (machine travel)		MPa
Yield strength (extensometer)		MPa
Ultimate tensile strength (UTS)		MPa
Young's modulus (machine travel)		GPa
Young's modulus (extensometer)		GPa

5.1 Calculations

Calculate the yield strength, ultimate tensile strength (UTS) and Young's modulus (E) for both datasets (machine travel and extensometer). Use the results to identify the unknown aluminium alloy.

5.2 Graphs / diagrams

[Insert diagram or figure here, with labelled axes and units. Plotting can be done in Python or Excel.]

6. Discussion

6.1 Interpretation of results

Discuss the difference in the linear elastic slope (E) between the two curves. Explain why the extensometer is required to get true material properties, and use the measured properties to determine the unknown aluminium alloy.

6.2 Error analysis

Systematic errors: *[Error source 1]*

Random errors: *[Error source 1]*

7. Conclusion

[Summarise the most important findings and answer the question posed in the objective.]

8. References

- [1] Häll, K. (2026). Materials Science & Engineering: Practical Characterization. University of South-Eastern Norway.
- [2] USN Department of Science and Industry Systems (2026). TSM2370 Materialteknikk 2026 course material. University of South-Eastern Norway.
- [3] USN Kongsberg Mechanical Engineering Lab (2026). Equipment guides (Lloyd LR 10K, Matsuzawa MXT50). github.com/ravo10/USN-kongsberg-lab-mech

9. Appendix

9.1 Raw data

[Raw data tables or measurement sheets]

9.2 Additional figures

[Further graphs, photos or sketches]

Date: _____ Signature: _____

I hereby confirm that this protocol was written independently and truthfully.

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Lab 2: Thermal Processing and Quench Severity

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Experiment date	*[DD/MM/YYYY]*
Lab partner(s)	*[Lab partner names]*

1. Objective

Measure the local resistance to plastic deformation (Vickers hardness) of different microstructures in Sølvstål 2210 (1.2210 / 115CrV3, a high-carbon tool steel). The experiment shows how the quench medium (cold water vs. warm water or oil) and the thermal processing route decide the final mechanical properties.

2. Theoretical background

Hardness correlates strongly with a material's yield strength. The experiment uses two separate thermal units:

- **Primary austenitizing furnace (810 °C):** Samples B, C and D are all first placed in the primary furnace at 810 °C, inside the hardening range given in the datasheets for 1.2210 (780–840 °C), and held for 15–30 minutes. The matrix transforms to austenite (γ, FCC) and most of the carbon dissolves in the iron lattice. Because this steel has about 1.15 % carbon, a small amount of carbide stays undissolved. That is normal for a high-carbon tool steel and gives wear resistance; heating much higher would cause grain growth and a higher risk of cracking.
- **Quenching (cold water vs. warm water or oil):** Samples B and C are cooled rapidly, directly from 810 °C. Rapid quenching prevents carbon diffusion and forces the FCC austenite into a body-centred tetragonal (BCT) lattice: martensite, the hardest and most brittle microstructure. A 20 °C cold-water quench (Sample B) is more severe than 70 °C warm water or oil (Sample C). 1.2210 is a shallow-hardening steel, so in the slower quench of Sample C part of the austenite may transform to softer pearlite or bainite before martensite can form. Expect Sample C to be clearly softer than Sample B.
- **Isothermal annealing and air cooling (600 °C oven → ~20 °C air):** Sample D is NOT quenched in liquid. After austenitizing at 810 °C it is moved directly to a secondary oven at 600 °C and held for 15–30 minutes. Because 600 °C is well above the martensite start temperature (Ms), carbon diffuses quickly and the austenite transforms fully into soft, ductile pearlite (ferrite + cementite). When the transformation is complete, the sample is taken out and air-cooled on a heat-resistant surface to room temperature (~20 °C) for handling and hardness testing.

2.1 Relevant formulas

Vickers hardness: $HV = 1.8544 \cdot F / d^2$

Mean diagonal: $d = (d_1 + d_2) / 2$

Units: ***F in kgf, d in mm. This lab uses HV1 (F = 1 kgf).***

3. Materials and equipment

Device / material	Amount / specification	Manufacturer / grade	Notes
High-carbon steel blocks	Prepared pieces (Sølvstål 2210)	1.2210 / 115CrV3	4 prepared samples per group
Lab furnaces and ovens	Primary furnace (810 °C) and secondary oven (600 °C)	–	Austenitizing (810 °C) and isothermal anneal (600 °C)
Quench media stations	20 °C water, 70 °C water, oil	–	Different quench severity
Vickers hardness tester	1 unit	Matsuzawa (MXT50)	HV1 (1 kgf), 15 s dwell
Polishing equipment	As needed	–	Prepare surfaces for indentation

4. Experimental procedure

4.1 Experimental setup

Four prepared Sølvstål 2210 blocks (A, B, C, D) are handed out. The primary furnace is preheated to 810 °C for austenitizing Samples B, C and D. The quench baths are brought to their temperatures (20 °C water and 70 °C warm water or oil). A secondary oven is preheated to 600 °C for the isothermal anneal of Sample D. Hardness is measured on the Matsuzawa MXT50; see the equipment guide at github.com/ravo10/USN-kongsberg-lab-mech.

4.2 Procedure

1. Take safety precautions and put on heat-resistant protective equipment (gloves, tongs, face shield). Keep the lid for the oil bath within reach.
2. **Sample A (baseline):** Measure the baseline Vickers hardness of the prepared, untreated steel block.
3. **Sample B (cold-water quench):** Place in the 810 °C primary furnace and hold for 15–30 minutes. Quench rapidly in 20 °C cold water, moving the sample around in the bath with tongs. Polish and measure the Vickers hardness.
4. **Sample C (reduced-severity quench):** Place in the 810 °C primary furnace and hold for 15–30 minutes. Quench in 70 °C warm water OR quenching oil. Polish and measure the Vickers hardness.
5. **Sample D (600 °C isothermal anneal and air cooling):** Place in the 810 °C primary furnace and hold for 15–30 minutes. Move it directly to the secondary oven at 600 °C and hold for 15–30 minutes for complete pearlite transformation. Take it out and let it air-cool on a heat-resistant surface to room temperature (~20 °C). Polish and measure the Vickers hardness.
6. Record all hardness indentations and clear the testing equipment.

5. Results and measurements

Parameter / measurement	Value	Unit
Sample A hardness (baseline, untreated)		HV1
Sample B hardness (810 °C → 20 °C water quench)		HV1
Sample C hardness (810 °C → 70 °C water / oil quench)		HV1
Sample D hardness (810 °C → 600 °C anneal → 20 °C air)		HV1

5.1 Calculations

Record the Vickers hardness of all four samples in the table. If you make several indentations per sample, give the mean value.

5.2 Graphs / diagrams

[Insert diagram or figure here, with labelled axes and units. Plotting can be done in Python or Excel.]

6. Discussion

6.1 Interpretation of results

Compare Sample B (cold water) and Sample C (warm water or oil) to explain quench severity and cooling rate in high-carbon tool steel, and whether Sample C hardened fully. Compare Sample B (martensite) and Sample D (810 °C → 600 °C for 15–30 min, then air-cooled) to explain how the cooling route controls the phase transformation (BCT martensite vs. pearlite) and the final hardness.

6.2 Error analysis

Systematic errors: *[Error source 1]*

Random errors: *[Error source 1]*

7. Conclusion

[Summarise the most important findings and answer the question posed in the objective.]

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